BRUNVOLL

Technical Report

Cable Lay Vessel – Azimuth vs Shaft Line Propulsion Comparison

Brunvoll Document No. XXXXXX(Pre)

September 29, 2026

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1 Introduction

This report compares two propulsion configurations for a 6000 t DP class 2 cable lay vessel (CLV), in operation and in transit. The two configurations are:

Unit	Option 1: Azimuth	Option 2: Shaft line
Main propulsion	2 x 3000 kW azimuth, D 3.3 m, ducted	2 x 3000 kW shaft line, D 3.0 m, ducted
Steering	(azimuth)	2 x flap rudder, 6.9 m^2
Stern tunnel thrusters	–	2 x 1500 kW, D 1.5 m
Bow tunnel thrusters	2 x 1700 kW, D 2.6 m	2 x 1700 kW, D 2.6 m
Bow retractable thruster	1 x 1600 kW, D 2.4 m	1 x 1600 kW, D 2.4 m

Both main propulsion options are modelled as **fixed pitch propellers with variable speed electric motors**.

The comparison covers:

1. Static DP capability (DNV-ST-0111 environment), intact and after worst case failure, for DP station keeping and for ploughing / cable lay.
2. Propulsion power in operation at fixed weather (Beaufort 3 and 6), intact.
3. Propulsion power in transit, 8–12 kn, calm water and Beaufort 5 head weather.

All calculations are made with the BruCon DP models (thrust allocation, vessel model, wave drift) and study drivers written for this comparison.

2 Models and assumptions

2.1 Vessel and power system

- Reference hull of an existing offshore vessel of similar size:

Main dimension	Value
Length between perpendiculars	101.1 m
Length overall	111.5 m
Breadth	22.4 m
Draft	6.5 m
Block coefficient	0.74
Displacement	about 11100 t
Lateral / frontal wind area	1755 / 700 m^2

- Wind forces by Blendermann's method (Blendermann 1994, 2013) with the coefficients for an offshore supply vessel (transverse drag 0.90, longitudinal drag 0.55 head / 0.80 stern, cross-force parameter 0.55), lateral and frontal wind areas as above.
- Wave drift forces from strip theory (PdStrip) response functions, mean drift by Newman's approximation. Current forces from the section based hull model.
- The same hull is used for both options, so the comparison isolates the propulsion. In reality a shaft line hull would have a somewhat finer stern.

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- Three-split power system (port / centre / starboard switchboard, bus ties), 4 x 3600 kW generators on port and starboard buses, 2.9 MW ESS on the centre bus. Identical for both options.
- Thrusters per bus:

Bus	Azimuth option	Shaft line option
Port	Bow tunnel 1, azimuth 1	Bow tunnel 1, stern tunnel 1, main prop 1
Centre	Bow retractable	Bow retractable
Starboard	Bow tunnel 2, azimuth 2	Bow tunnel 2, stern tunnel 2, main prop 2

2.2 Thrusters (DP and ploughing)

Table 2-1 lists the thrusters with the assumed net bollard thrust used in the DP and ploughing calculations, and their positions: x from midship ($L_{pp}/2$), positive forward; y from the centre line, positive to starboard.

Table 2-1: Thrusters, assumed net bollard thrust and positions. T_+: ahead (tunnels: to starboard). T_-: astern (tunnels: to port). Azimuths give T_+ in any direction.

Option	Unit	Type	D [m]	P [kW]	T_+ [kN]	T_- [kN]	x [m]	y [m]
Both	Bow tunnel 1	Tunnel	2.6	1700	244	244	42.8	0.0
Both	Bow tunnel 2	Tunnel	2.6	1700	244	244	38.6	0.0
Both	Bow retractable	Azimuth, ducted	2.4	1600	307	–	34.0	0.0
Azimuth	Port azimuth	Azimuth, ducted	3.3	3000	459	–	-48.1	-5.4
Azimuth	Stbd azimuth	Azimuth, ducted	3.3	3000	459	–	-48.1	5.4
Shaft line	Stern tunnel 1	Tunnel	1.5	1500	101	101	-38.7	0.0
Shaft line	Stern tunnel 2	Tunnel	1.5	1500	101	101	-41.5	0.0
Shaft line	Port main propeller	FPP, ducted	3.0	3000	459	275	-46.0	-4.5
Shaft line	Stbd main propeller	FPP, ducted	3.0	3000	459	275	-46.0	4.5
Shaft line	Port / stbd rudder	Flap, 6.9 m^2	–	–	–	–	-48.5	$\mp$ 4.5

- Main propulsors: simple fixed pitch thrust / power model at low speed. Net bollard thrust 459 kN per unit at 3000 kW for **both** options. This assumes the ship designer has aimed for equal bollard pull, the larger azimuth propeller compensating for gear housing, strut and hull interaction losses. For the shaft line the value is the D-series fixed pitch bollard design (540 kN gross) times $(1 - t)$ with t = 0.15.
- Astern: shaft line reversing at 60 per cent of ahead thrust (to be confirmed). The azimuths are not reversed; they rotate.
- Azimuth thruster–thruster forbidden sectors of 60 deg (wash onto the other azimuth).
- Stern tunnels: 35 per cent thrust loss due to stern tube blockage, 101 kN per unit (30–40 per cent is typical; the result is insensitive within that range).
- Bow tunnels and retractable: estimates scaled from reference thrusters by power and diameter.
- Rudder side force from the BruCon rudder / slipstream model (flap rudders, 45 deg max angle, 3.2 m height).
- Thrust bias (azimuth pair pushing against each other) off, unless stated.

2.3 Environment (DNV-ST-0111)

- Wind, wave and current relation from DNV-ST-0111 Table 2-1, collinear. Beaufort values between the table points are interpolated linearly in wind speed.

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- Wave drift from a Pierson–Moskowitz (Bretschneider) spectrum with $\cos^2$ spreading (Section 3.3.3), T_p as tabulated.
- Dynamic allowance 1.25 on the environmental forces in the capability analysis. Operational loads (tow force, own-speed resistance) are not factored. No dynamic allowance in the power calculations (mean power).
- The analysis is not a formal ST-0111 Level 1 rating: the thruster modelling and thrust allocation are vessel specific (BruCon allocator).

2.4 Operation modes and failure cases

- **DP station keeping**: zero speed.
- **Ploughing** / **cable lay**: 0.8 kn ahead through the water and an 80 t (785 kN) pull acting astern on the centre line.
- **Worst case failure (WCF)**: loss of one switchboard (port, centre or starboard). The thrusters on that bus are lost, the remaining buses run as separate groups limited by their own generators. The WCF curve is, per heading, the lowest of the three bus losses.

3 DP capability

Figure 3-1 shows the maximum mean wind speed that can be held as a function of the weather direction relative to the bow, with the Beaufort upper limits from DNV-ST-0111 as rings. Table 3-1 summarises the results. The search is limited to 40 m/s.

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Figure 3-1: DP capability, maximum mean wind speed held versus weather direction (0 deg = head weather). Top: DP station keeping. Bottom: ploughing / cable lay, 0.8 kn and 80 t pull. Left: intact. Right: worst case failure.

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Table 3-1: DP capability, wind speed in m/s (interpolated Beaufort in brackets). Stern sector = lowest value with weather from 120–240 deg.

Mode	Case	Azimuth min	Shaft line min	Azimuth stern sector	Shaft line stern sector
Station keep.	Intact	18.9 (7.5)	18.2 (7.3)	24.1 (8.9)	18.5 (7.4)
Station keep.	WCF	14.6 (6.2)	13.6 (5.9)	18.3 (7.3)	13.6 (5.9)
Ploughing	Intact	14.9 (6.3)	14.9 (6.3)	23.3 (8.7)	22.6 (8.5)
Ploughing	WCF	not feasible	not feasible	–	–

Observations:

- The lowest capability is similar for both options. For the azimuth option it is set by the bow thrusters at about 60 deg off the bow; for the shaft line option by the stern side force with weather aft of the beam.
- The main difference is with weather from the stern quarters (120–240 deg). Intact, the azimuth option holds about 5 m/s more wind, and after WCF about 4.7 m/s more (Bf 7.3 against 5.9). After a bus loss the azimuth option keeps a full azimuth at the stern, while the shaft line option has one propeller with rudder and one small, partly blocked stern tunnel.
- Ploughing intact is limited by the forward thrust needed for the 80 t pull and is equal for both options (equal bollard pull by assumption).
- Ploughing after WCF is not feasible for either option: after a bus loss about 78 t is available in calm water. In practice the operation would be stopped after a WCF, and the focus is station keeping while ending the operation, so this case mainly illustrates the missing thrust reserve.

4 Power consumption in operation

Figure 4-1 shows the propulsion power (propeller shaft power, all thrusters) at fixed weather, Beaufort 3 and 6, as a function of the weather direction. The azimuth option is shown with thrust bias off and on. Table 4-1 gives the mean over all headings.

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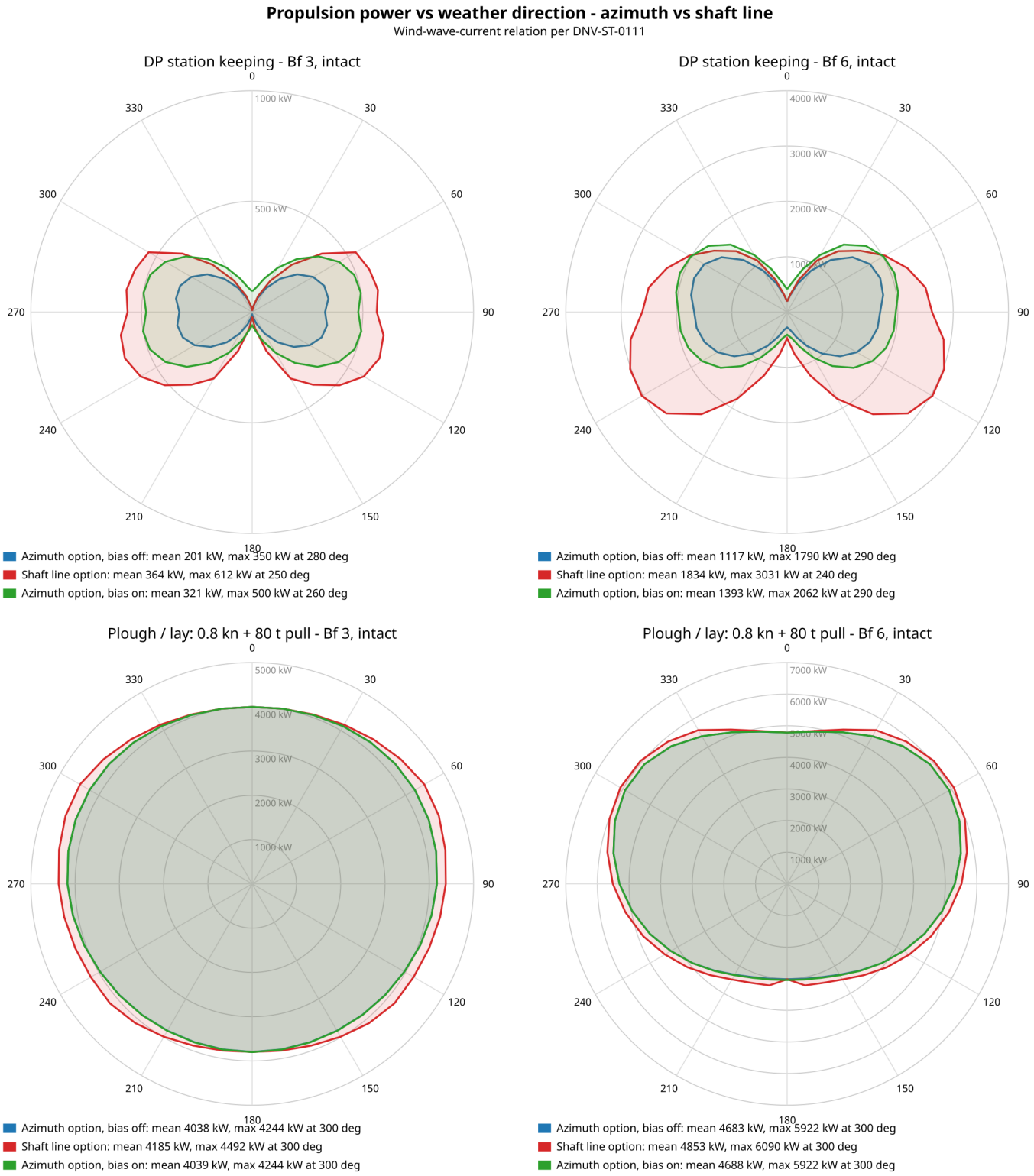


Figure 4-1: Propulsion power versus weather direction, intact. Top: DP station keeping. Bottom: ploughing / cable lay. Left: Beaufort 3. Right: Beaufort 6.

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Table 4-1: Propulsion power, intact, mean over all headings (max in brackets), kW.

Mode	Weather	Azimuth, bias off	Azimuth, bias on	Shaft line
Station keep.	Bf 3	201 (350)	321 (500)	364 (612)
Station keep.	Bf 6	1117 (1790)	1393 (2062)	1834 (3031)
Ploughing	Bf 3	4038 (4244)	4039 (4244)	4185 (4492)
Ploughing	Bf 6	4683 (5922)	4688 (5922)	4853 (6090)

Observations:

- Station keeping: little power is needed with head or stern weather for either option; beam and quartering weather dominate. The shaft line option needs clearly more power with weather aft of the beam, where it must combine propeller, rudder and the small stern tunnels. At Bf 6 the azimuth option uses about 40 per cent less power with bias off and about 24 per cent less with bias on.
- Thrust bias costs the azimuth option 120–280 kW, but it stays below the shaft line.
- Ploughing: the 80 t pull dominates. The azimuth option uses 3–4 per cent less, as the azimuths can take the side forces by angling the thrust while pushing.
- Motor, converter and gear losses are not included, and the thrust / power model is a simple relation. The relative differences are more reliable than absolute values.

5 Transit power

The transit power is calculated with a separate model: calm water resistance (ITTC-57 friction, form factor 1.3, Holtrop–Mennen wave making), appendage and tunnel opening drag, and for Beaufort 5 head weather the wind force (Blendermann) and wave drift (strip theory). The main propellers are modelled as ducted fixed pitch propellers (D-series at design pitch), with two design philosophies, as the ship designer has not specified this:

- **A, bollard design**: pitch for maximum bollard thrust at 3000 kW; in transit the propeller is light, and motor overspeed (20 per cent) is allowed.
- **B, transit design**: pitch for maximum speed at 3000 kW (calm water + 15 per cent margin); torque limited at bollard, about 10 per cent lower bollard thrust.

Coefficient	Azimuth	Shaft line
Wake fraction w	0.10	0.20
Thrust deduction t	0.10	0.15
Transmission efficiency	0.97 (angle gears)	0.99
Appendages	struts, gear housings	rudders, shafts, brackets, tunnels

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Table 5-1: Transit, total propulsion motor power for both units, kW.

Condition	Speed	Azimuth A	Azimuth B	Shaft line A	Shaft line B
Calm	8 kn	608	569	534	511
Calm	10 kn	1182	1105	1037	991
Calm	12 kn	2106	1972	1859	1779
Bf 5 head	8 kn	1207	1166	1166	1152
Bf 5 head	10 kn	2006	1924	1906	1870
Bf 5 head	12 kn	3203	3063	3015	2947

Observations:

- In calm water the shaft line option needs about 10–12 per cent less power; in Bf 5 head weather the difference is 1–5 per cent.
- Most of the calm water difference comes from the assumed hull efficiency $(1-t)/(1-w)$ (about 6 per cent) and gear efficiency (about 2 per cent). These are estimates; the difference is of the same order as the uncertainty in the inputs.
- The transit design (B) saves 4–7 per cent in calm water compared with the bollard design (A), at the cost of about 10 per cent bollard thrust. Top speed is about 16 kn for all variants.

6 Conclusions

- With equal bollard pull, the two options are close in the headline numbers: lowest DP capability, ploughing capability and transit power are within about half a Beaufort number or 10 per cent.
- The main difference is the **stern-wise capability**: with weather from the stern quarters the azimuth option holds about 5 m/s more wind, both intact and after worst case failure (after WCF Bf 7.3 against 5.9). For a CLV, which has to follow the cable route and cannot freely choose heading, this gives a larger operating window.
- The azimuth option uses less power in DP station keeping (about 25–40 per cent at Bf 6) and slightly less in ploughing (3–4 per cent).
- The shaft line option needs about 10 per cent less power in calm water transit, less in weather; this depends mainly on uncertain propulsive coefficients.
- Effects not captured by a static analysis, such as faster thrust response, no dependence of side force on propeller thrust, and thrust losses of the stern tunnels in current, all favour the azimuth option in practice.
- Whether the stern-wise capability and lower DP power outweigh the transit power, cost and maintenance aspects is a question for the operator.

7 Open points and inputs needed

- Net bollard thrust of the azimuths including gear housing, strut and hull interaction losses (here set equal to the shaft line).
- Wake fraction, thrust deduction and gear efficiency for both options.

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- Astern thrust of the reversing fixed pitch shaft line propellers (here 60 per cent), and four-quadrant data for ducted fixed pitch propellers.
- Stern tunnel thrust including stern tube blockage.
- Hull lines and strip theory data of the actual vessel (reference hull used here).
- Operating profile (time in transit / DP / ploughing) and site weather statistics, to express the difference as operability and annual energy.

8 Notes on the calculation tools

During the study an error was found in the BruCon thrust allocation: for a twin main propeller / rudder vessel with one main propeller lost, the side force given to the remaining propeller and rudder was not realised when a bow azimuth was present. This is fixed in BruCon issue 13044 (branch `13044-dp-allocation-for-twin-main-...`), and the results in this report are made with the fix.

The configurations, study drivers, plot script and results are stored in `hydro_tools/cablelay_propulsion`, see `README.md` there for how to reproduce.

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